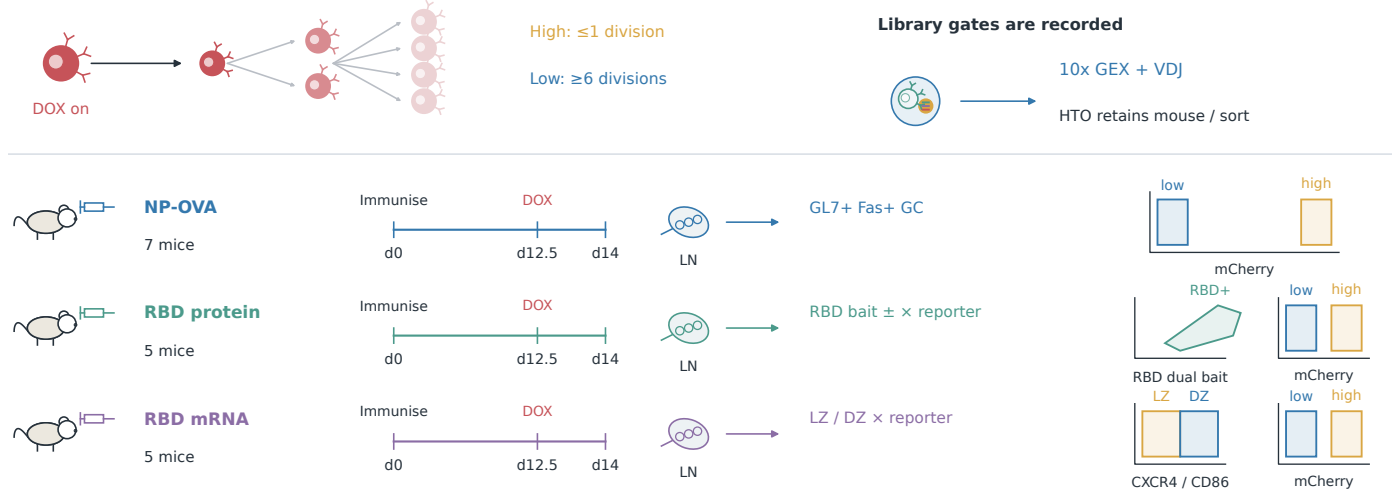


Figure 2 | Interpreting clone state in experimentally measured GC selection

NP-OVA and RBD models: clone position is state similarity, not ancestry or a future fate.

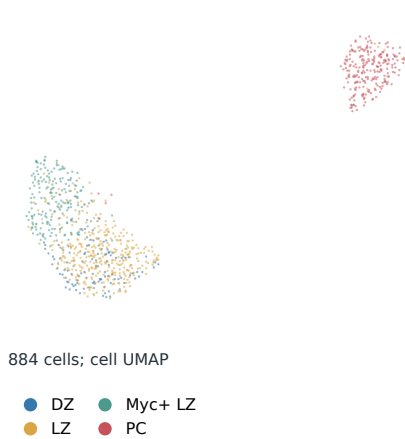
A Reporter mechanism and experimental gates retain the original biological design

H2B-mCherry reporter: stop synthesis, read 36 h of division history

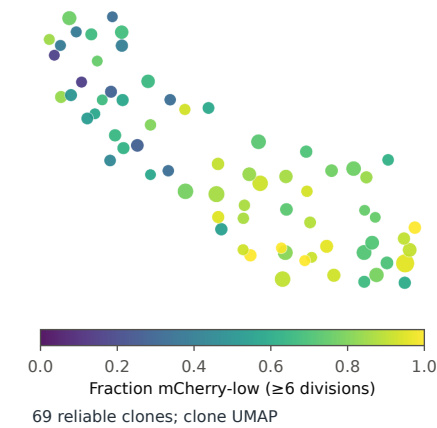


Independent NP, protein and mRNA cohorts; gate outlines are schematic, not measured FACS events.

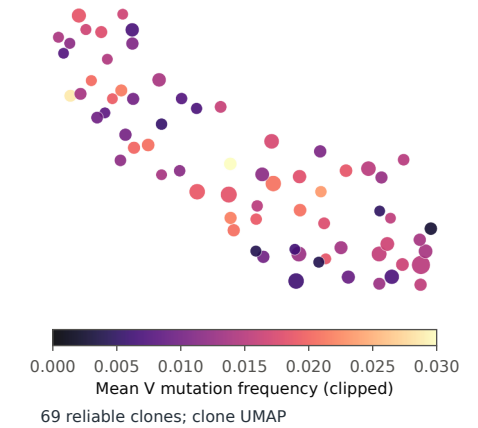
B NP-OVA d14: measured GC and output gates



C Which clones favour more divisions?



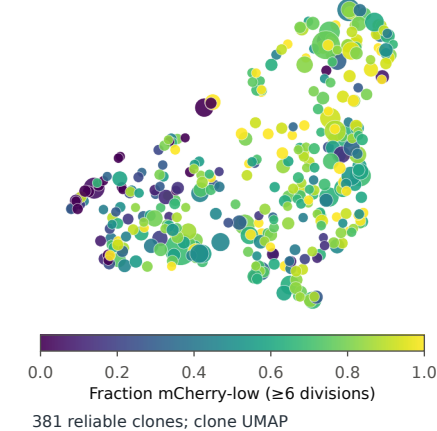
D Does mutation load explain the same map?



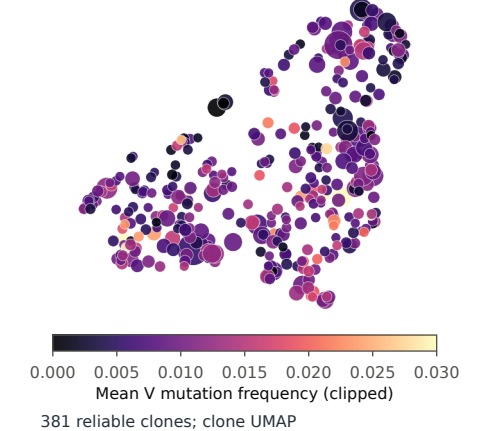
E RBD vaccines: cell division gate



F Which clones favour more divisions?



G Does mutation load explain the same map?



H How to read the three views

B: GSE246382, independent NP-OVA output-sort study (Smart-seq2 / TRUST4); C/D: NP-OVA division reporter. E-G: RBD reporter cohorts. Clone maps compare state distributions; their coordinates are not lineage trees.